

# **PRE-SYNOPSIS REPORT**

on

## **Integrating Multi-Source Data with Sentiment Analysis and Language Models to Enhance Stock Market Decision Making**

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## Abstract

Equity markets are driven simultaneously by structured price and volume records, by qualitative information carried in news and social media, and by slower macroeconomic conditions. Each stream is individually incomplete and each becomes informative or uninformative at different times. Conventional forecasting systems compress this heterogeneity into a single numerical pipeline, report a statistical accuracy figure, and stop short of the decision the investor actually has to take. This research addresses that shortfall by developing, implementing and validating an integrated framework in which multi-source fusion, language-model-based sentiment extraction, transparent reasoning and decision-level allocation are treated as one connected engineering problem rather than as four separate ones.

The first objective develops a hierarchical Multi-Source Fusion Network combining convolutional indicator encoders, transformer text encoders, engineered macroeconomic features and a cross-modal attention layer that assigns time-varying weight to each source. On ten years of National Stock Exchange data across ten sectoral indices it attains a normalised mean absolute error of 0.018,  $R^2$  of 0.988 and directional accuracy of 84.7 per cent at the index level, against 76.3 per cent for a simple averaging ensemble over identical inputs; removing both textual sources costs 10.9 per cent of explained variance, exceeding the sum of their individual removals. The fusion rule is then strengthened by a Bayesian formulation in which the weight on a source is derived from its own recent predictive uncertainty through a softmax over negative error variances, improving the Sharpe ratio by 22.7 per cent and reducing maximum drawdown by 18.3 per cent against the strongest static-fusion baseline, while the same experts fused with fixed weights add nothing — which locates the gain in the weighting mechanism rather than in the networks.

The second objective couples a transformer sentiment encoder with fifteen technical indicators under a strict temporal protocol. Language-model sentiment emerges as the highest-gain predictor, correlating 0.74 with momentum yet only 0.04 with price, and raises directional accuracy from a 46.43 per cent baseline mean to 47.72 per cent while producing the highest prediction confidence of all models examined. The absolute level is reported without embellishment: at single-name daily horizons all models cluster near the random classifier. The third objective replaces correlation-based feature selection with a directed causal adjacency matrix recovered by conditional independence testing, encodes intraday, daily and weekly horizons through a multi-scale memory encoder, and allocates capital through a credibilistic optimiser. Over 308 NIFTY50 sessions it recovers a causal graph of density 0.850 with an identifiable causal hub and shows that strong causal influence can coexist with weak linear correlation; the sentiment composite carries the expected sign against four independently published series and separates the identified market states at  $p = 1.8 \times 10^{-8}$ , yet is separately shown not to forecast returns.

The fourth objective evaluates the framework as a decision system. Over eighteen years of ten liquid multi-asset instruments under a walk-forward protocol with transaction costs it attains a Sharpe ratio of 0.878 against 0.628 for an equal-weighted benchmark and reduces maximum drawdown from  $-36.1$  to  $-19.0$  per cent, winning in all four stress episodes, in 18 of 19 calendar years on drawdown and in all 18 specifications examined, and beating all three benchmarks on a 665-session holdout that postdates every design decision. Ablation at matched average exposure isolates the source of the gain: timing risk by market state improves both the Sharpe ratio and the drawdown with no change in mean return, whereas the signal fusion layer contributes nothing detectable. Cross-market transfer reproduces the defensive behaviour but loses on risk-adjusted return, bounding applicability to universes with genuine dispersion in asset volatility. Taken together, the six publications underlying this report demonstrate a coherent progression from information integration, through language-model-based signal extraction and transparent reasoning, to validated decision making; the component-level nulls are retained because they identify precisely which layers of a fusion architecture earn their place and which do not.

**Keywords:** multi-source data fusion; sentiment analysis; large language models; explainable artificial intelligence; regime detection; portfolio construction; Indian equity market.

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## Abbreviations

|         |                                          |         |                                                     |
|---------|------------------------------------------|---------|-----------------------------------------------------|
| ANOVA   | Analysis of Variance                     | MAE     | Mean Absolute Error                                 |
| ARIMA   | Autoregressive Integrated Moving Average | MDD     | Maximum Drawdown                                    |
| AUC     | Area Under the Curve                     | ML      | Machine Learning                                    |
| Bi-LSTM | Bidirectional Long Short-Term Memory     | MLP     | Multi-Layer Perceptron                              |
| BSE     | Bombay Stock Exchange                    | MOSPI   | Ministry of Statistics and Programme Implementation |
| CAGR    | Compound Annual Growth Rate              | MSFN    | Multi-Source Fusion Network                         |
| CIT     | Conditional Independence Test            | NLP     | Natural Language Processing                         |
| CNN     | Convolutional Neural Network             | NSE     | National Stock Exchange                             |
| CPI     | Consumer Price Index                     | OHLCV   | Open, High, Low, Close, Volume                      |
| CVaR    | Conditional Value at Risk                | QP      | Quadratic Program                                   |
| DA      | Directional Accuracy                     | RBI     | Reserve Bank of India                               |
| DCHF    | Dynamic-Causal Hybrid Framework          | RDM-DEA | Range Directional Measure Data Envelopment Analysis |
| DL      | Deep Learning                            | RMSE    | Root Mean Square Error                              |
| ETF     | Exchange-Traded Fund                     | RSI     | Relative Strength Index                             |
| GMM     | Gaussian Mixture Model                   | SEBI    | Securities and Exchange Board of India              |
| GRU     | Gated Recurrent Unit                     | SHAP    | Shapley Additive Explanations                       |
| IC      | Information Coefficient                  | SMA     | Simple Moving Average                               |
| IIP     | Index of Industrial Production           | SVM     | Support Vector Machine                              |
| LLM     | Large Language Model                     | VADER   | Valence Aware Dictionary and sEntiment Reasoner     |
| LSTM    | Long Short-Term Memory                   | VIX     | Volatility Index                                    |
| MACD    | Moving Average Convergence Divergence    | WPI     | Wholesale Price Index                               |
|         |                                          | XAI     | Explainable Artificial Intelligence                 |

## Chapter 1: Introduction

The price of a traded security is the outcome of a continuous negotiation between participants who do not observe the same information at the same time. Part of that information is structured and numerically clean — open, high, low, close and volume records and the technical indicators derived from them. A larger part is unstructured and arrives irregularly, in news articles, corporate disclosures, regulatory announcements and social media commentary. A third part changes slowly and shapes the environment rather than the individual security: policy rates, inflation, industrial output and currency movements. Any forecasting system that reads only one of these streams is, by construction, reasoning from an incomplete description of the market it is trying to anticipate [1, 2].

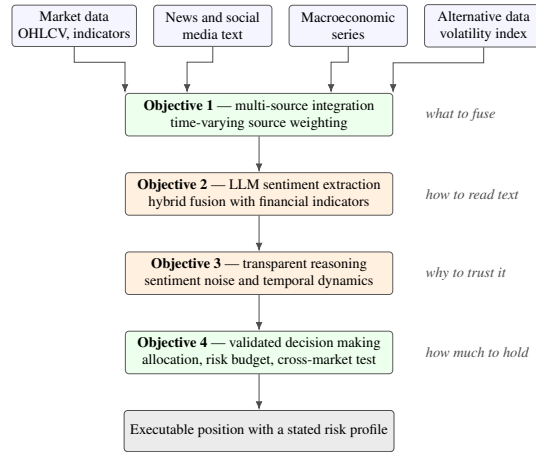
Two developments have made the integration of these streams practical rather than aspirational: the maturation of multi-source data fusion, which supplies a principled account of how heterogeneous streams may be combined so that their complementary content is preserved instead of being averaged away [1, 3, 4]; and the arrival of transformer-based large language models, which read financial text with a contextual sensitivity that lexicon and bag-of-words methods never achieved and can therefore convert narrative information into a numerically usable signal [48, 45, 36]. Together they make it reasonable to ask whether a single system can absorb quantitative, textual and macroeconomic evidence and convert it into a decision an investor could actually execute.

That question is sharper than it appears, because three difficulties sit between a fused signal and a defensible decision. First, the reliability of a source is not constant: technical indicators are informative in quiet trending conditions and degrade when the market is disturbed, while news and volatility signals behave in the opposite manner, so a fusion rule with fixed weights misweights its inputs precisely when accurate weighting matters most [5, 11]. Second, a prediction produced by a deep architecture is not by itself an account of why the prediction was made; where capital is committed on the strength of the output, opacity is a practical obstacle and not merely a philosophical one [56, 57]. Third, predictive accuracy and economic usefulness are different properties: a forecast becomes an investment only after it has been reconciled with a covariance structure, a risk budget, position limits and the cost of trading, and that translation can destroy the value of a forecast as easily as preserve it [83, 94].

This research is directed at those three difficulties simultaneously. Its title, *Integrating Multi-Source Data with Sentiment Analysis and Language Models to Enhance Stock Market Decision Making*, states the ambition, and the emphasis falls on the final two words. The work is not an attempt to add another architecture to a crowded field of predictors, but to build, instrument and honestly evaluate the complete path from heterogeneous raw information to an executable allocation, and to determine which stages of that path actually contribute to the outcome. Figure 1.1 places the four research objectives on that path: Objective 1 governs the integration of sources, Objective 2 the extraction of signal from language and its combination with conventional indicators, Objective 3 the transparency of the reasoning and the behaviour of sentiment as a noisy, time-lagged input, and Objective 4 the conversion of a signal into a position and the validation of that conversion across markets. The objectives are sequential in that each consumes the output of the one before it, but they are not merely stages of a pipeline: each raises a distinct research question and each is answered by distinct experimental evidence.

The Indian equity market is the primary empirical setting, by choice rather than convenience. The National Stock Exchange combines high retail participation, strong sensitivity to domestic policy events such as the Union Budget and Reserve Bank of India monetary policy announcements, pronounced sectoral concentration in financial services, and simultaneous exposure to global factors including crude oil prices and United States monetary policy [8, 12, 35]. These features make it an unusually demanding test of a multi-source system, because the sources genuinely disagree and the disagreement is informative. Cross-market validation on United States multi-asset and mega-capitalisation universes is carried out separately under Objective 4 to establish where the framework transfers and where it does not.

The remainder of this report is organised as follows. Chapter 2 reviews the literature and states the research gaps that follow from it. Chapter 3 formalises the problem statement and the four objectives. Chapter 4



**Figure 1.1:** Research context: the path from heterogeneous market information to an executable decision, and the position of the four research objectives on that path.

states the research contributions and maps the candidate’s published work onto those objectives. Chapter 5 presents the methodology, implementation and experimental evidence for each objective and shows how the contributions compose into a single system. Chapter 6 discusses the consolidated results and the significance of the work. Chapter 7 concludes and states the future scope, and Chapter 8 gives the intended organisation of the thesis.

## Chapter 2: Literature Survey

This chapter reviews the research directly relevant to the thesis along the same four axes as the objectives, so that the gap identified at the end of each section is the gap the corresponding objective addresses. Emphasis is placed on what each body of work establishes and, more importantly, on what it leaves unresolved.

### 2.1 Multi-Source Data Fusion in Financial Prediction

That heterogeneous sources outperform any single source is now well supported. Snášel et al. [1] formalised a generalised multi-source fusion framework for stock selection and showed that complementary streams mitigate each other’s individual weaknesses; Long et al. [2] reached the same conclusion from a multi-view heterogeneous data perspective; and Nti et al. [4] reported accuracy gains of 15–20 per cent from combining technical, fundamental and sentiment features while emphasising that sources sampled at different frequencies must be synchronised before they are combined. The pattern recurs across investor-sentiment models [8], multi-source sentiment and price fusion [27], trend forecasting [26], multi-level dynamic fusion [28], calendar-aware recurrent modelling [29], multiple instance learning over text and price [31], long-sequence transformers over social media [32], multi-task prediction of correlated financial variables [33] and emerging-market recurrent models with exogenous variables [34]. Parallel work on single-source architectures establishes the ceiling that fusion must beat: Song [24] enriched the feature space for trend-reversal prediction and Javaid et al. [25] combined a peephole recurrent network with a time-attention layer, both improving on conventional deep baselines while remaining confined to price-derived inputs.

The methodological question is not whether to fuse but how, and three strategies dominate. Early fusion concatenates features from all sources before modelling, which is simple but forces one architecture to learn cross-domain interactions and is prone to modality dominance [6, 30]. Late fusion trains source-specific models and combines their predictions, preserving modularity but usually fixing the combination weights in advance [5, 17]. Hybrid or attention-based fusion learns modality importance inside the network [21, 9, 7, 75, 11], a genuine advance that nonetheless produces weights which are point estimates over an entire training set rather than statements about the reliability of a source at a particular moment. A survey of thirty-six recent multi-source studies conducted as part of this research found that 44.4 per cent combine text with technical data and 25.0 per cent remain price-only, but that *none* conditions its fusion weights on a time-varying uncertainty estimate. Where fusion output is carried through to a portfolio at all, the conversion is typically heuristic: Sami et al. [20] construct holdings by statistical averaging of forecasts, without a covariance structure or an explicit risk objective.

### 2.2 Sentiment Analysis and Language Models in Finance

Financial text analysis has passed through three generations. Dictionary methods scored text against pre-defined polarity lexicons; Loughran and McDonald [47] showed that general-purpose lexicons misclassify financial language badly enough to require domain-specific alternatives, and VADER [46] remains a widely used reference implementation. Supervised machine learning improved accuracy but required large annotated corpora and generalised poorly across market conditions [23]. Transformer architectures [48] displaced both: FinBERT [45] adapted bidirectional encoders to financial language, and domain models [42, 43, 44] extended this to generative reasoning over financial documents. Surveys converge on the finding that contextual language models extract market-relevant meaning that lexicon methods miss [36, 37, 38].

Applications have followed quickly. Conversational language models outperform established financial sentiment baselines [40]; dedicated models exist for sentiment in financial tweets [41]; Haryono et al. [9, 10] coupled a transformer news encoder with a gated recurrent unit and later extended it to generative aspect-based sentiment; Ho and Huang [22] combined sentiment with candlestick chart representations; Jung and Jang [13] improved sentiment quality by masking numerically significant tokens so that quantitative con-

text is preserved; and Ónózó et al. [15] applied language models to news analysis and macroeconomic nowcasting. Applied studies span markets and asset classes, and the obstacles are surveyed [39, 17, 18, 19, 72, 73]. What this literature establishes is that language models improve sentiment *measurement*. What it establishes far less clearly is how much of that improvement survives into forecast quality once the sentiment score is fused with price history, aligned to trading sessions and evaluated without look-ahead.

### 2.3 Explainability, Causality and Uncertainty

Opacity is a recognised obstacle to the adoption of learned models in finance. Koa et al. [56] used self-reflective large language models to generate natural-language explanations alongside stock predictions, and general-purpose attribution methods [57, 58] are now standard instruments for post-hoc interpretation. Post-hoc attribution, however, explains a model; it does not establish that the relationships the model has learned are anything more than correlations that happen to hold in the training window.

Causal formulations address this directly. Xu et al. [59] showed that causal feature selection is more robust out of sample than correlation-based feature engineering, but their causal graph is static and cannot represent the structural breaks produced by policy announcements. Liu et al. [60] introduced a dynamic-causal lens that encodes stock interactions as symbolic movement patterns and adapts to changing conditions, at a computational cost that scales quadratically in the number of instruments. Related work couples factor investing with causal graph networks using factors calibrated to United States equities [61], adds memory to a causal interaction network [62], or models the vagueness of inter-stock relations [65]. Alongside this, multi-granularity architectures [64, 63] demonstrated that financial series carry structure at several time scales at once, with Wang et al. [63] adding a debiasing layer that assumes stationary noise properties — an assumption in tension with the conditional heteroskedasticity that has been the standard description of financial volatility since Bollerslev [90].

Relational formulations extend the same idea to the cross-section [66, 67, 68], treating securities as nodes in a learned graph and modelling their mutual interplay. The statistical instrument underlying directed inference here remains the notion of predictive precedence formalised by Granger [101] and the constraint-based discovery procedures of Spirtes et al. [102], while the learning components rest on long short-term memory [49], gated recurrent units [50] and gradient-boosted tree ensembles [51].

A related and less examined issue is that a signal’s reliability is itself uncertain. Most fusion architectures pass point predictions forward without any statement of confidence, so a source that has recently become noisy continues to be weighted as though it had not; Bacco et al. [12] study prediction explicitly under high uncertainty yet still fuse sentiment and price statically. Explicit, time-varying uncertainty quantification [103, 104] is the natural remedy and is largely absent from the reviewed fusion literature.

### 2.4 From Prediction to Decision: Allocation and Validation

Portfolio construction has its own long literature. Markowitz [79] established the mean-variance formulation, whose extreme sensitivity to estimated inputs motivated both shrinkage estimators [82] and the demonstration by DeMiguel et al. [83] that naive equal weighting can outperform optimisation that ignores estimation error. Regime-switching models following Hamilton [85] showed that financial series are not homogeneous through time; Ang and Bekaert [86] demonstrated that accounting for regime shifts materially changes optimal allocations, and Kritzman et al. [87] made the practical case for regime-aware dynamic strategies. Volatility-managed portfolios [89] scale total risk by recent volatility but say nothing about composition; Kumar and Yadav [69] handle positive and negative returns without assuming normality but without integration to a dynamic signal; risk-based construction [84] forgoes signal information by design; and recent work generates risk-aware prediction intervals [70] or builds sparse portfolios from irregular, English-only news streams [71]. Position sizing under an estimated edge remains governed by the growth-optimal criterion of Kelly [80] and the information-ratio scaling of Grinold and Kahn [92].

A second and older family of sentiment measures reads sentiment off the market rather than off text. Baker and Wurgler [91] construct investor sentiment from trading activity and dispersion, on the argument that participation reveals what surveys merely report, and later work builds allocation signals on exactly this

reasoning [77, 78]. This family matters here because instrument-level news corpora do not exist for broad index instruments, which constrains what can be measured at the allocation layer.

Evaluation practice is the final concern. Risk-adjusted performance is still summarised by the ratio introduced by Sharpe [81], and Gu et al. [94] provide the benchmark machine-learning study in asset pricing, noting that measured predictability is small. Harvey and Liu [95] showed that conventional significance thresholds are inadequate once the number of strategies examined is taken into account, and Bailey and López de Prado [96] formalised the deflated Sharpe ratio for exactly that reason. Robust inference under serial dependence requires corrections [98], resampling schemes [99] and forecast-comparison tests [100]. The practical consequence is that a reported gain is credible only when the specification search behind it is reported alongside it.

Table 2.1 consolidates the representative work reviewed above, stating for each the contribution that is relevant to this thesis and the limitation this thesis must work around.

**Table 2.1:** Consolidated review of representative work, with the limitation relevant to this research

| Reference                                      | Technique                                                 | Contribution relevant here                                                                     | Limitation to be addressed                                                                          |
|------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Snášel et al. [1]                              | Generalised multi-source fusion for stock selection       | Formal basis for combining numerical and textual streams                                       | Fusion weights are not conditioned on the prevailing market state                                   |
| Long et al. [2];<br>Nti et al. [4]             | Multi-view and multi-source deep fusion                   | Views carry non-overlapping information; frequency alignment is essential                      | Developed and tested outside emerging-market conditions                                             |
| Haryono et al. [9];<br>Mu et al. [8]           | Sentiment fused with technical indicators                 | Gated and optimised fusion of text and price streams                                           | Gate learned globally; sentiment quality assumed rather than validated                              |
| Farimani et al. [5]                            | Adaptive multimodal weighting                             | Moves toward time-varying source importance                                                    | No explicit probabilistic confidence per source                                                     |
| Araci [45]; Wu et al. [42]                     | Domain-adapted language models                            | Contextual sentiment extraction from financial text                                            | Gain in sentiment quality is not traced through to decision quality                                 |
| Koa et al. [56];<br>Lundberg and Lee [57]      | LLM explanations; post-hoc attribution                    | Natural-language and feature-level justification                                               | Explains the model, not the causal structure                                                        |
| Xu et al. [59]; Liu et al. [60]                | Causal and dynamic-causal graph modelling                 | Causal features are more robust out of sample                                                  | Static graph cannot absorb policy shocks; dynamic variant scales quadratically                      |
| Wang et al. [63]                               | Multi-scale temporal memory with debiasing                | Multiple horizons processed jointly                                                            | Debiasing assumes stationary noise                                                                  |
| Kumar and Yadav [69];<br>Moreira and Muir [89] | Credibilistic selection; volatility-managed portfolios    | Optimisation without normality; total risk scaled by recent volatility                         | Neither is coupled to a dynamic signal, and neither addresses composition and risk level separately |
| DeMiguel et al. [83]; Harvey and Liu [95]      | Naive diversification; backtesting under multiple testing | Estimation error can defeat optimisation; reported gains must account for specification search | Sets the benchmark any framework must beat and requires every specification to be disclosed         |

## 2.5 Research Gaps

The review supports five gaps, stated below and traced to objectives in Figure 3.1.

- G1. Static fusion of dynamically reliable sources.** Existing frameworks fuse financial, textual and macroeconomic streams using fixed weights, globally learned attention or simple averaging. None conditions the contribution of a source on a measured, time-varying estimate of that source’s own predictive reliability, so a stream that has become noisy continues to influence the forecast at its historical weight.
- G2. Incomplete coverage of the Indian information environment.** Most multi-source frameworks are designed and validated on developed markets. Frameworks that jointly incorporate financial indicators, news sentiment, social media signals and Indian macroeconomic variables — policy rate, infla-

tion, industrial production, currency — under a single architecture and a strict temporal protocol are rare.

- G3. Unverified translation from language-model sentiment to decision quality.** Language models demonstrably improve sentiment measurement. Whether that improvement survives temporal alignment, fusion with technical indicators and honest out-of-sample evaluation at a single-name horizon has not been established, and reported gains are seldom accompanied by the baseline against which they should be judged.
- G4. Correlational rather than mechanistic transparency.** Post-hoc attribution explains a fitted model; it does not distinguish a genuine driver from a spurious association, nor describe how a sentiment shock propagates into a price movement over time. A mechanism grounded in directed causal structure and multi-scale temporal dynamics is required, together with a direct measurement of how much noise the sentiment channel carries.
- G5. Prediction reported in place of decision.** Statistical accuracy remains the dominant success criterion and economic value is reported inconsistently. Where allocation is addressed it is often heuristic, and the contribution of individual layers is almost never isolated, so architectures accumulate components whose value has never been tested. Cross-market transfer is asserted more often than demonstrated.

## Chapter 3: Problem Statement and Research Objectives

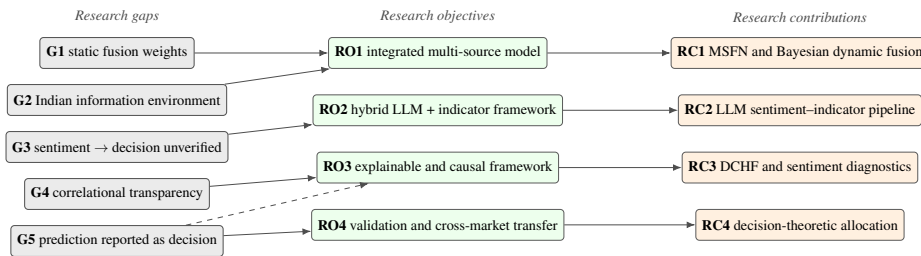
### 3.1 Problem Statement

To design, develop and validate an integrated decision-support framework that fuses heterogeneous financial information — market metrics, news and social media sentiment extracted using large language models, and macroeconomic indicators — into interpretable and reliability-aware predictions, and that converts those predictions into executable, risk-controlled stock market decisions whose performance is demonstrated across market states and across markets. The framework must address the limitations of existing approaches, namely static source weighting under non-stationary reliability, opaque and correlational reasoning, unquantified sentiment noise, and the routine substitution of statistical accuracy for economic value.

### 3.2 Research Objectives

- RO1.** To investigate and develop a comprehensive model that integrates diverse data sources — such as financial metrics, news articles, social media sentiment and macroeconomic factors — aiming to enhance the accuracy of stock market predictions by providing a holistic analysis.
- RO2.** To design a hybrid framework that combines large language models, sentiment analysis techniques and conventional financial indicators, focusing on optimising deep learning architectures for real-time stock market predictions.
- RO3.** To establish and evaluate an explainable artificial intelligence framework capable of delivering transparent and interpretable predictions in stock selection, analysing the impact of sentiment data noise and exploring the temporal dynamics between sentiment shifts and stock price movements.
- RO4.** To evaluate and validate the result and practical application of the proposed model across different stock markets, exploring transfer learning techniques for cross-market applications and validating the model against benchmark datasets and real-world market data.

Figure 3.1 traces each gap to the objective that addresses it and to the contribution reported in Chapter 4. The mapping is many-to-many by design: G1 and G2 are jointly answered by the fusion work of RO1, while G5 requires evidence from both the decision layer of RO4 and the transparency instruments of RO3.



**Figure 3.1:** Alignment of research gaps, research objectives and research contributions. Dashed links denote partial rather than primary coverage.

## Chapter 4: Research Contributions

This chapter states what has been contributed against each research objective and identifies the published work in which that contribution appears. Each publication is described once, at the point where its contribution is actually made; later chapters refer to it by its label [P1]–[P6], with the methodology and evidence developed in Chapter 5 and full bibliographic details in *Publications on PhD Work*.

### 4.1 Research Contribution 1 — Integrated multi-source model (RO1)

A comprehensive multi-source model was designed, implemented and evaluated for the Indian equity market. The Multi-Source Fusion Network of [P1] integrates four streams — technical indicators from National Stock Exchange price and volume records, news sentiment from a domain-adapted transformer encoder, social media sentiment from a hybrid lexicon and recurrent classifier, and Indian macroeconomic series — through modality-specific encoders, a cross-modal attention layer assigning time-varying relevance, and a bidirectional recurrent encoder, validated on ten years of data across ten sectoral indices under a strictly chronological split with walk-forward confirmation. The contribution is not that fusion improves accuracy, which was already known, but that the improvement is *attributable*: ablation isolates the marginal value of each stream, and the attention distribution shows source importance shifting with the market regime.

That observation exposed a limitation in the fusion mechanism itself, addressed in [P4]. A Bayesian dynamic uncertainty-weighted framework models each source with its own expert network, estimates each expert’s predictive uncertainty continuously from the variance of its recent errors, and obtains the fusion weights by a softmax over the negative uncertainties, so that a source is discounted at the moment its predictions become unreliable rather than at the next retraining cycle. This is the methodological core of the objective: it converts fusion from a question of *what* to combine into a question of *how* to combine under changing reliability.

### 4.2 Research Contribution 2 — Hybrid LLM and indicator framework (RO2)

A hybrid framework combining large language models with conventional financial indicators was designed and evaluated in [P3]. News headlines are read by a transformer sentiment encoder producing a continuous polarity score rather than a categorical label, preserving intensity that binary classification discards; the score is time-aligned to trading sessions with explicit handling of closing hours and publication timestamps, then fused at feature level with fifteen technical indicators and passed to a gradient-boosted ensemble, against technical-only baselines under temporal cross-validation.

Three findings constitute the contribution. The sentiment score is the highest-gain predictor in the fused model, ahead of established technical features, so language-model sentiment carries information price history does not; its low correlation with price-based features confirms that the information is orthogonal rather than redundant; and — reported without softening — the incremental gain in directional accuracy is small, with all models close to the random classifier at a single-name daily horizon. The framework’s practical value lies instead in the highest prediction confidence of the models examined and the strongest performance on buy signals. Delivering it in a reproducible, freely accessible environment was an explicit design goal, so the result can be independently re-derived.

### 4.3 Research Contribution 3 — Explainable and causal framework (RO3)

Transparency was addressed mechanistically rather than by post-hoc attribution alone. The Dynamic-Causal Hybrid Framework of [P2] replaces correlation-based feature selection with a directed causal adjacency matrix recovered by conditional independence testing, so that the features entering the model are those with demonstrable directed influence rather than those that merely co-move; temporal structure is captured by a multi-scale memory encoder at intraday, daily and weekly granularity, and the signal is converted into holdings by a credibilistic optimiser. Applied to NIFTY50 constituent data over 308 trading sessions it recovers an interpretable causal graph, identifies a systematic causal hub, and demonstrates

that strong directed causal influence can exist between securities whose linear correlation is weak — the justification for the approach, since correlational selection would have discarded a genuine driver.

The second half of the objective, sentiment noise and sentiment-to-price temporal dynamics, is addressed by [P3] and [P5]. The former measures accuracy separately under positive, neutral and negative sentiment regimes and shows that sentiment spikes frequently precede price movements while the mapping from magnitude to direction is not simple. The latter treats the sentiment construct as a testable claim rather than an assumption: validated against four independently published series, the composite carries the expected sign against every one in every sub-period and separates the identified market states at  $1.8 \times 10^{-8}$  — and the same study then measures separately whether it forecasts anything, and reports that it does not. Keeping construct validity and predictive usefulness apart is itself a contribution.

#### 4.4 Research Contribution 4 — Validated decision framework and cross-market transfer (RO4)

A decision-theoretic framework converting a fused signal into an executable portfolio was specified and evaluated in [P5]. Latent market states are identified by a Gaussian mixture refitted at every rebalance with explicit label anchoring; composites are fused with state-dependent weights and scaled onto a return metric by information-ratio scaling; the scaled signal enters a constrained quadratic program with covariance shrinkage, position caps and a turnover penalty; and the sleeve is scaled by a regime-dependent risk budget. The design deliberately separates *what to hold relative to everything else* from *how much total risk to carry*, so that each layer can be removed and tested independently. Evaluation spans eighteen years of daily data on ten multi-asset instruments with transaction costs, a fully reported specification sweep, four stress episodes selected on documented market history, a 665-session holdout postdating every design decision, and a secondary mega-capitalisation universe as a cross-market test.

The contribution has three parts: the framework itself, specified in enough detail to be reproduced from the description alone; an ablation design isolating component contributions *at matched average exposure*, necessary because scaling a portfolio toward cash changes return and volatility together and would otherwise confound risk timing with risk level; and, most useful to a practitioner, the identification of two failure modes — an absolute volatility target never reached on a diversified universe, which silently disables the regime layer, and a signal-to-allocation conversion in which mismatched units rather than information determine the optimiser’s solution. Both were found only by disassembling the framework and both generalise beyond it. The cross-market component is answered honestly: transferred unchanged, the framework retains its defensive behaviour but loses on risk-adjusted return — a bounded claim, which is more useful than an unbounded one.

#### 4.5 Consolidated Mapping of Published Work to Objectives

Table 4.1 states the relationship between each publication and each objective. A *primary* mapping means the publication was designed to answer the objective and supplies its principal evidence; a *partial* mapping means it contributes a specific, identifiable component without answering the objective as a whole. Blank cells are deliberate and should be read as an accurate statement of scope rather than as an omission. The structure of the research programme is visible in the table: two publications feed the integration layer, one supplies the language-model signal, one supplies transparency and one supplies the decision layer, and the evidence accumulates in sequence rather than in parallel. [P6] is comparative groundwork that established the gap the later work addresses.

**Table 4.1:** Mapping of published work to the research objectives, with the nature of the contribution in each case

| Ref. | Publication (short title)                              | RO1      | RO2      | RO3      | RO4      | Nature of the contribution                                                                         |
|------|--------------------------------------------------------|----------|----------|----------|----------|----------------------------------------------------------------------------------------------------|
| [P1] | Multi-source data fusion framework (conference)        | <b>P</b> | p        | —        | p        | Four-stream architecture, cross-modal attention, source ablation, ten-sector evaluation            |
| [P2] | Dynamic-causal hybrid framework, NIFTY50 (conference)  | p        | —        | <b>P</b> | p        | Directed causal feature selection, multi-granularity memory, credibilistic allocation              |
| [P3] | LLM-extracted news sentiment and market data (journal) | p        | <b>P</b> | p        | —        | LLM sentiment pipeline, feature-level fusion, sentiment-regime accuracy analysis                   |
| [P4] | Bayesian dynamic uncertainty-weighted fusion (journal) | <b>P</b> | p        | p        | —        | Time-varying source reliability, softmax weighting over negative uncertainties                     |
| [P5] | Signal fusion to asset allocation (journal)            | —        | —        | p        | <b>P</b> | Regime detection, constrained allocation, matched-exposure ablation, holdout and cross-market test |
| [P6] | Deep learning in stock forecasting (conference)        | p        | p        | —        | —        | Comparative groundwork that established the research gap                                           |

**P** = primary contribution; **p** = partial contribution; **—** = no contribution claimed.

## Chapter 5: Research Methodology and Proposed System Implementation

This chapter develops the methodology objective by objective. For each objective it states the design, describes the implementation as an ordered engineering procedure, presents the experimental evidence, and records the publication in which the work was validated. Section 5.6 then shows how the four components compose into a single system.

### 5.1 Overall Research Framework and Workflow

The framework is organised as the four layers of Figure 1.1. Each layer consumes the output of the previous one and each corresponds to one research objective, so that a defect in any layer can be located rather than absorbed into an aggregate performance figure. Layer 0 acquires four heterogeneous streams and aligns them to a common trading calendar without look-ahead. Layer 1 (RO1) encodes each modality and fuses them under cross-modal attention and then under measured source reliability. Layer 2 (RO2) applies the optimised deep architecture that turns the fused representation into a prediction. Layer 3 (RO3) supplies the transparency instruments — directed causal structure and sentiment diagnostics — and returns a reliability revision to the fusion layer, so that a source found to be noisy is down-weighted rather than merely reported. Layer 4 (RO4) converts the signal into an allocation under risk, cost and turnover constraints and validates it by walk-forward testing, a specification sweep, matched-exposure ablation, a holdout and a second market. The sequence was executed once end to end and then revisited iteratively as each objective produced evidence that constrained the next.

Two deliberate design decisions govern the data. The study universes differ by objective because the objectives ask different questions: sectoral breadth is required to test fusion (NIFTY 50 and ten sectoral indices, January 2015 to December 2024, with fifteen technical indicators, financial press, X and StockTwits, and RBI, MOSPI and IMF macroeconomic series); single-name depth is required to test sentiment (Indian large-capitalisation equities with daily OHLCV and news headlines scored by a transformer); a compact cross-sectional universe is required to recover causal structure (five NIFTY50 constituents across five sectors, 308 sessions from January 2024); and cross-asset volatility dispersion is required to test allocation (ten multi-asset instruments over 4,689 sessions from January 2008, with a secondary mega-capitalisation universe). And the evaluation window lengthens as the objectives progress, from a ten-year sectoral study to an eighteen-year multi-asset study with a dedicated holdout, because claims about behaviour under market stress require observations of market stress. Each study uses the protocol appropriate to its question — chronological splitting with training-set-only standardisation and expanding-window walk-forward for RO1 and RO2, conditional independence testing for RO3, and 21-session walk-forward with 10 bp transaction costs, an 18-cell specification sweep, matched-exposure ablation and a 665-session holdout for RO4.

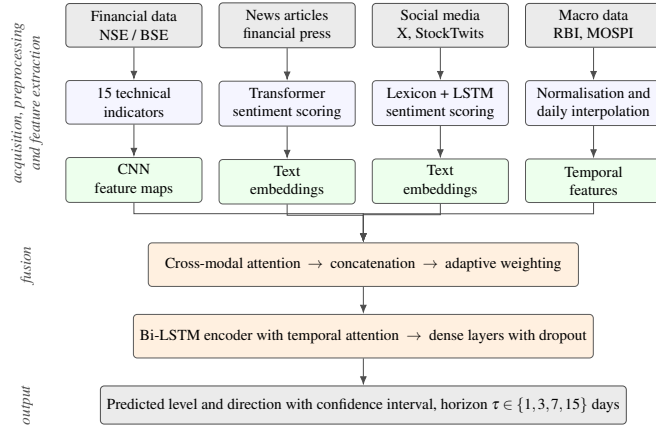
### 5.2 Proposed Methodology for Research Objective 1

**RO1:** *To investigate and develop a comprehensive model that integrates diverse data sources — financial metrics, news articles, social media sentiment and macroeconomic factors — aiming to enhance the accuracy of stock market predictions.*

#### 5.2.1 Design and implementation of the multi-source fusion network

The Multi-Source Fusion Network is a hierarchical architecture that learns modality-specific representations before learning cross-modal ones, so that each stream is encoded in a form appropriate to its own structure (Figure 5.1). **(1) Acquire and align:** collect price and volume records, news articles, social media posts and macroeconomic releases and align every stream to the trading calendar; monthly and quarterly macroeconomic series are interpolated to daily frequency, and the consequence — an understatement of discontinuous announcement-day effects — is stated explicitly rather than left implicit. **(2) Engineer modality features:** derive fifteen technical indicators covering trend, momentum, volatility and volume; score news articles with a domain-adapted transformer encoder weighted by relevance and source authority; score social media posts with a hybrid lexicon and fine-tuned recurrent classifier,

retaining daily polarity, mention volume and dispersion; normalise the macroeconomic series. **(3) Encode each modality** by one-dimensional convolution, with  $h_m^{(l)}$  the representation of modality  $m$  at layer  $l$ :  $h_m^{(l)} = \text{ReLU}(\text{Conv1D}(W_m^{(l)} * h_m^{(l-1)} + b_m^{(l)}))$ . **(4) Fuse by cross-modal attention**: compute a relevance weight  $\alpha_m = \text{softmax}(q \cdot k_m / \sqrt{d_k})$  per modality, where  $q$  is a learned query vector,  $k_m$  the key vector of modality  $m$  and  $d_k$  the key dimension; the weights are recomputed at each time step, which is what allows source importance to move with the market state. **(5) Model temporal dependence**: pass the fused representation  $F_t$  through bidirectional recurrent layers,  $h_t = [\vec{h}_t; \overleftarrow{h}_t]$  with  $\vec{h}_t = \text{LSTM}(F_t, \vec{h}_{t-1})$  and  $\overleftarrow{h}_t = \text{LSTM}(F_t, \overleftarrow{h}_{t+1})$ , and produce forecasts at horizons  $\tau \in \{1, 3, 7, 15\}$  days through dense layers with dropout [54], trained by adaptive moment estimation [55] with early stopping. **(6) Evaluate without leakage**: partition chronologically into training (2015–2021), validation (2022 – mid-2023) and test (mid-2023 – 2024) blocks, standardise using training-set statistics only, tune hyperparameters on the validation block alone, and fix random seeds so that reported differences reflect architecture rather than initialisation.



**Figure 5.1:** Architecture of the Multi-Source Fusion Network, showing the flow from heterogeneous acquisition through modality-specific preprocessing and feature extraction to cross-modal fusion and prediction.

## 5.2.2 Results and discussion for Research Objective 1

Table 5.1 reports the comparison against six baselines together with the source ablation, presented as one table because the two questions — does fusion help, and which source is responsible — are answered by the same experiment.

**Table 5.1:** One-day-ahead prediction on the NIFTY 50 index: comparison against baselines (upper block) and source ablation of the proposed model (lower block)

| Configuration                                | Data sources                | MAE          | RMSE         | $R^2$        | DA (%)      |
|----------------------------------------------|-----------------------------|--------------|--------------|--------------|-------------|
| <i>Baseline comparison</i>                   |                             |              |              |              |             |
| ARIMA                                        | price only                  | 0.124        | 0.167        | 0.721        | 52.3        |
| LSTM                                         | price only                  | 0.087        | 0.112        | 0.845        | 61.7        |
| Transformer                                  | price only                  | 0.079        | 0.103        | 0.869        | 64.2        |
| LSTM + sentiment                             | price, sentiment            | 0.052        | 0.071        | 0.921        | 72.8        |
| LSTM + macro                                 | price, macro                | 0.061        | 0.082        | 0.903        | 68.5        |
| Ensemble (simple average)                    | all, fixed weights          | 0.043        | 0.059        | 0.942        | 76.3        |
| <b>MSFN (proposed)</b>                       | <b>all, adaptive fusion</b> | <b>0.018</b> | <b>0.024</b> | <b>0.988</b> | <b>84.7</b> |
| <i>Source ablation of the proposed model</i> |                             |              |              |              |             |
| – financial metrics                          | news, social, macro         | 0.041        | 0.055        | 0.946        | 73.2        |
| – news sentiment                             | price, social, macro        | 0.027        | 0.036        | 0.971        | 79.1        |
| – social media                               | price, news, macro          | 0.025        | 0.033        | 0.976        | 80.8        |
| – macroeconomic                              | price, news, social         | 0.031        | 0.040        | 0.965        | 76.5        |
| – both sentiment sources                     | price, macro                | 0.038        | 0.049        | 0.952        | 74.0        |
| price and technical only                     | price only                  | 0.079        | 0.103        | 0.869        | 64.2        |

Three readings follow. First, adaptive fusion is materially better than fixed-weight combination: the simple

averaging ensemble uses exactly the same four sources and attains a mean absolute error of 0.043 against 0.018, which isolates the contribution of the weighting rule rather than of the data. Second, financial metrics remain foundational — their removal costs 22.6 per cent of explained variance — but they are not sufficient, since restricting the model to price and technical features costs 32.0 percentage points of directional accuracy. Third, the joint removal of both textual sources costs 10.9 per cent of explained variance, exceeding the sum of their individual removals of 5.3 and 3.6 per cent; that superadditivity is the quantitative signature of complementary rather than redundant information, and it is the empirical justification for a fusion architecture instead of a larger single-source model.

The learned attention distribution explains where the gain comes from. Financial metrics carry the highest average weight at 35 per cent, rising to 42 per cent during earnings season, when company-specific numerical disclosure dominates, and falling to 28 per cent under high volatility, when the social media weight rises from 18 to 28 per cent; news sentiment averages 22 per cent and macroeconomic indicators 25 per cent, so the textual sources jointly carry 40 per cent. This is precisely the behaviour a fixed-weight scheme cannot reproduce, and it is consistent with the finding that social media sentiment influences Indian sectoral returns most strongly at market extremes [35]. Because the headline accuracy is high, the possibility that it is an artefact of a fortunate split was tested under an expanding-window protocol in which each fold is evaluated only on sessions strictly later than its training window. Normalised errors are tightly clustered at  $0.0016 \pm 0.0004$  and directional accuracy remains above chance in every fold, ranging from 68 to 92 per cent with a mean of 78.4 per cent; fold-level  $R^2$  averages 0.874, naturally lower than the full-test value because each short window spans a narrow price range, and the stability of the error magnitudes, not the level of  $R^2$ , is the relevant evidence. Sector-wise evaluation adds a second dimension: financials and fast-moving consumer goods are the most predictable at  $R^2$  above 0.985 while metals and real estate fall below 0.962, and sentiment contributes most to the globally exposed information technology and pharmaceutical sectors whereas energy and metals depend more heavily on macroeconomic indicators. Source importance is therefore structured by sector as well as by regime.

### 5.2.3 Strengthening the fusion rule: dynamic uncertainty weighting

Attention weights are learned once over a training set and express *average relevance* rather than *instantaneous reliability*: a source that has become unreliable in the last two weeks continues to receive its historical weight. To remove that limitation the fusion rule was reformulated on a Bayesian basis. Each source  $i \in \{M, S, A\}$  — market series, sentiment sequence, volatility index — is modelled by an expert network matched to its structure: a gated recurrent unit, a transformer encoder and a multi-layer perceptron respectively. The predictive uncertainty  $\sigma_{i,t}^2$  is estimated as the variance of that expert’s prediction errors over a trailing window of  $\tau = 10$  sessions, and the uncertainties are converted into fusion weights by a softmax over their negatives, so that the fused forecast  $\hat{y}_t = \sum_i w_{i,t} \hat{y}_{i,t}$  uses

$$w_{i,t} = \frac{\exp(-\sigma_{i,t}^2/T)}{\sum_j \exp(-\sigma_{j,t}^2/T)}, \quad (5.1)$$

with  $T$  a temperature parameter: a source whose recent error variance has risen is automatically down-weighted. Distribution-aware normalisation is applied throughout — standard scaling for returns, min-max for sentiment, robust scaling for the heavy-tailed volatility index — fitted on an expanding window so that no future information enters the scaler.

The mechanism self-corrects in a way a static scheme cannot: if a sentiment score is distorted by an ambiguous or ironic headline and the subsequent market move contradicts it, the error enters  $\sigma_{S,t}^2$  and the sentiment weight falls over the following sessions without manual intervention. This is the direct methodological answer to gap G1. The formulation was evaluated on a heterogeneous Nifty 50 dataset spanning January 2018 to December 2023: index price and volume records with fifteen technical indicators, a daily sentiment score from a language model fine-tuned on Indian financial news, and the India VIX as an alternative source of forward-looking volatility. News sentiment is assigned to the next National Stock Exchange session so that lagged market impact is captured without look-ahead, market holidays are treated

as genuine gaps rather than forward-filled, and the split is strictly sequential — 2018–2021 training, 2022 validation, 2023 out-of-sample test — with the experts pre-trained independently and the system then fine-tuned end to end through the fusion layer. The comparison is against the three static fusion strategies identified in Chapter 2: early fusion, in which all sources are concatenated into one recurrent model and the weights are fixed by the architecture [6]; late fusion, in which the three expert predictions are averaged with pre-determined weights [5]; and attention fusion, in which static neural attention over modalities is learned once and then held fixed [7, 9] — the strongest of the three. A fourth configuration, the proposed architecture with the uncertainty term disabled, isolates the weighting rule from the expert networks. The source study reports improvements relative to the strongest baseline rather than absolute levels.

Two readings follow. Against the strongest static competitor the dynamic rule improves the Sharpe ratio by 22.7 per cent and reduces maximum drawdown by 18.3 per cent, so the gain is in risk-adjusted terms and downside control rather than in raw accuracy. The second reading matters more: the ablation row shows that the same three expert networks, fused with fixed weights, perform no better than the established baselines. The experts are therefore not the source of the improvement; the weighting rule is — which allows the contribution to be stated precisely as a fusion-mechanism result rather than an architecture result, and is the same style of component-level attribution that Objective 4 later applies to the allocation layer. The behaviour is interpretable in the way the design intends: during the volatility episodes in the test window the weight on the technical expert falls as its recent error variance rises and migrates to the volatility-index expert and, less strongly, to the sentiment expert, returning in calmer trending stretches. Because the shift is driven by measured error variance rather than by a learned prior, no regime label is required for it to occur.

#### 5.2.4 Summary and validation of Research Objective 1

The objective is achieved. A four-stream model evaluated on ten years of Indian market data across ten sectors attains MAE 0.018,  $R^2$  0.988 and directional accuracy 84.7 per cent against 76.3 per cent for a fixed-weight ensemble over identical inputs; ablation demonstrates that textual sources contribute super-additively and the attention analysis that source importance is regime-dependent. The fusion rule was then strengthened from learned average relevance to measured instantaneous reliability, closing the static-weighting gap. [P1] supplies the architecture, the ten-sector evaluation, the ablation and the walk-forward analysis; [P4] supplies the uncertainty formulation and the dynamic weighting mechanism.

### 5.3 Proposed Methodology for Research Objective 2

**RO2:** *To design a hybrid framework that combines large language models, sentiment analysis techniques and conventional financial indicators, focusing on optimising deep learning architectures for real-time stock market predictions.*

#### 5.3.1 Design and implementation of the hybrid LLM–indicator framework

The framework is organised as five interconnected layers — data collection, technical indicators, sentiment analysis, feature fusion, and model training with evaluation. Its design priority differs from that of Objective 1: rather than maximising the number of sources, it isolates one source — news text read by a transformer language model — and measures precisely what that source adds to a conventional indicator set. Daily price and volume records are retrieved for the target equities and financial news headlines are harvested with non-financial content filtered out and credible sources prioritised; publication timestamps are synchronised with session times so that news released after the close is attributed to the next trading day. Fifteen indicators are derived, covering trend (moving averages over 5, 10 and 20 sessions), momentum (relative strength index [105], moving average convergence divergence with signal line), volatility (Bollinger band measures, historical volatility) and participation (volume-weighted average price derivatives, volume momentum). A transformer sentiment encoder scores each headline, retaining a continuous polarity score  $s_t \in [-1, +1]$  rather than a class label, together with a confidence score, with batched inference so that the pipeline remains tractable in a constrained computational environment. Fusion is at feature level: technical indicators are standardised by  $z$ -score to equalise scales across measurement units,

sentiment is min–max scaled to preserve its interpretable range, and the two are concatenated into a single input vector. A gradient-boosted tree ensemble [51] is trained on the fused space, chosen for its tolerance of heterogeneous features and nonlinear interactions, alongside Random Forest [52], Support Vector Machine [53] and Logistic Regression baselines on technical indicators alone, so that the incremental value of sentiment is measured against a like protocol. Evaluation partitions chronologically at 80/20, applies temporal cross-validation, tests performance differences by Diebold–Mariano statistics [100], and converts predictive accuracy into economic terms through a simulated trading strategy.

### 5.3.2 Results and discussion for Research Objective 2

Table 5.2 reports the comparison. The result must be read carefully, because the headline number and the useful number are not the same.

**Table 5.2:** Single-name directional classification with and without LLM-extracted sentiment features

| Model                             | Feature set               | Accuracy      | Precision | Recall | F1     |
|-----------------------------------|---------------------------|---------------|-----------|--------|--------|
| Random Forest                     | technical only            | 0.4982        | 0.4345    | 0.6033 | 0.5052 |
| Support Vector Machine            | technical only            | 0.4246        | 0.4246    | 1.0000 | 0.5961 |
| Logistic Regression               | technical only            | 0.4702        | 0.4257    | 0.7107 | 0.5325 |
| XGBoost with sentiment            | technical + LLM sentiment | 0.4772        | 0.4103    | 0.5289 | 0.4621 |
| Mean of technical-only baselines  |                           | 0.4643        | —         | —      | —      |
| Improvement of the proposed model |                           | <b>+2.77%</b> | —         | —      | —      |

The sentiment-augmented model attains 47.72 per cent accuracy against a baseline mean of 46.43 per cent, a relative improvement of 2.77 per cent. It does not attain the highest raw accuracy: Random Forest reaches 49.82 per cent. Reporting only the first of these numbers would misrepresent the finding, so both are stated. What makes the result meaningful is the structure beneath the aggregate. On **feature importance**, the sentiment score is the highest-gain predictor in the fused model, ranking above established technical features such as the twenty-session moving average and the moving average convergence divergence; it is not a marginal addition the model tolerates but the feature it relies on most. On **orthogonality**, its correlation with price is 0.04 while its correlation with momentum indicators reaches 0.74, so sentiment aligns with momentum, as expected if news and price respond to the same events, yet carries information essentially independent of price level — the condition under which fusion can help at all. Across **sentiment regimes**, accuracy is higher under positive sentiment for all models, consistent with the asymmetric market response to adverse news, and the sentiment-augmented model varies least across regimes: a stability property rather than an accuracy property, and the basis of the economic result. In **economic terms**, despite comparable statistical accuracy it produces the highest mean prediction confidence and higher simulated cumulative returns over a thirty-session window, with success rates on buy signals near 72 and 65 per cent against the baselines. That divergence is itself the finding: a model that is right no more often, but is right with better-calibrated confidence, produces better decisions. The **limit** is stated plainly: receiver operating characteristic analysis places all models close to the random classifier line. Language-model sentiment does not overturn market efficiency at this horizon, and this report does not claim that it does.

### 5.3.3 Summary and validation of Research Objective 2

The framework establishes that language-model sentiment is the highest-importance feature in the fused space, that it is orthogonal to price-based features, and that it improves prediction confidence and simulated economic performance while delivering only a small increment in raw directional accuracy. The bounded nature of that increment is reported rather than concealed, and it motivates the shift of emphasis in Objective 4 from statistical accuracy to decision quality. Feasibility at a daily decision horizon is established by the optimised architecture of **[P1]**, whose Bi-LSTM encoder with attention converges in under two hours on a single consumer-grade accelerator with about 2.1 million trainable parameters and millisecond forward-pass latency. **[P3]** supplies the sentiment pipeline, the feature-level fusion, the feature-importance and sentiment-regime analysis, and the simulated trading evaluation.

## 5.4 Proposed Methodology for Research Objective 3

**RO3:** *To establish and evaluate an explainable artificial intelligence framework capable of delivering transparent and interpretable predictions in stock selection, analysing the impact of sentiment data noise and exploring the temporal dynamics between sentiment shifts and stock price movements.*

### 5.4.1 Design of the causal and multi-granularity framework

Transparency is pursued here at the level of structure rather than of attribution. A post-hoc explanation states which inputs a fitted model used; it does not establish that those inputs are drivers. The Dynamic-Causal Hybrid Framework therefore constrains the model to features with demonstrable directed influence before any learning takes place, in three stages — causal discovery, multi-granularity temporal encoding and credibilistic optimisation — implemented as follows. Conditional independence testing [102, 101] over lagged closing-price series constructs a time-varying adjacency matrix  $A(t)$ , retaining a directed edge  $e_{ij}$  only where series  $X_i$  improves the prediction of  $X_j$  beyond  $X_j$ 's own history. Parallel recurrent encoders then run over short (1-day), medium (5-day) and long (21-day) windows and are combined through a temporal attention and debiasing layer,  $h_t = \text{Debias}(\text{Conv}_{1 \times k}[h_{t-1}^{\text{intraday}}, h_{t-1}^{\text{daily}}, h_{t-1}^{\text{weekly}}])$ , so that volatility clustering at one scale does not obscure momentum at another. Expected returns are expressed as triangular fuzzy numbers, admitting vagueness in the market view rather than forcing a point estimate the data do not support, and weights are obtained by maximising entropy-normalised expected return net of conditional value at risk,  $\max_w(\mathbb{E}[\tilde{R}] - \lambda \text{CVaR}_\alpha(\tilde{R}))/\text{Entropy}(w)$ , within the range directional measure feasible set. The result is backtested against a passive benchmark and attributed to identifiable mechanisms rather than to the aggregate.

### 5.4.2 Results: causal structure as an explanation mechanism

Table 5.3 gives the recovered adjacency matrix over 308 trading sessions from January 2024 to March 2025.

**Table 5.3:** Dynamic causal adjacency matrix  $A(t)$  over five NIFTY50 constituents. Entry  $(i, j)$  is the strength of the directed influence of row  $i$  on column  $j$ .

| Influencing ↓ / Influenced → | HDFCBANK     | NESTLEIND | RELIANCE | TCS          | TITAN        |
|------------------------------|--------------|-----------|----------|--------------|--------------|
| HDFCBANK                     | 1.000        | 0.299     | 0.791    | 0.079        | 0.460        |
| NESTLEIND                    | 0.072        | 1.000     | 0.356    | 0.130        | 0.006        |
| RELIANCE                     | <b>0.913</b> | 0.324     | 1.000    | <b>0.650</b> | <b>0.849</b> |
| TCS                          | 0.110        | 0.670     | 0.516    | 1.000        | 0.436        |
| TITAN                        | 0.337        | 0.058     | 0.216    | 0.610        | 1.000        |

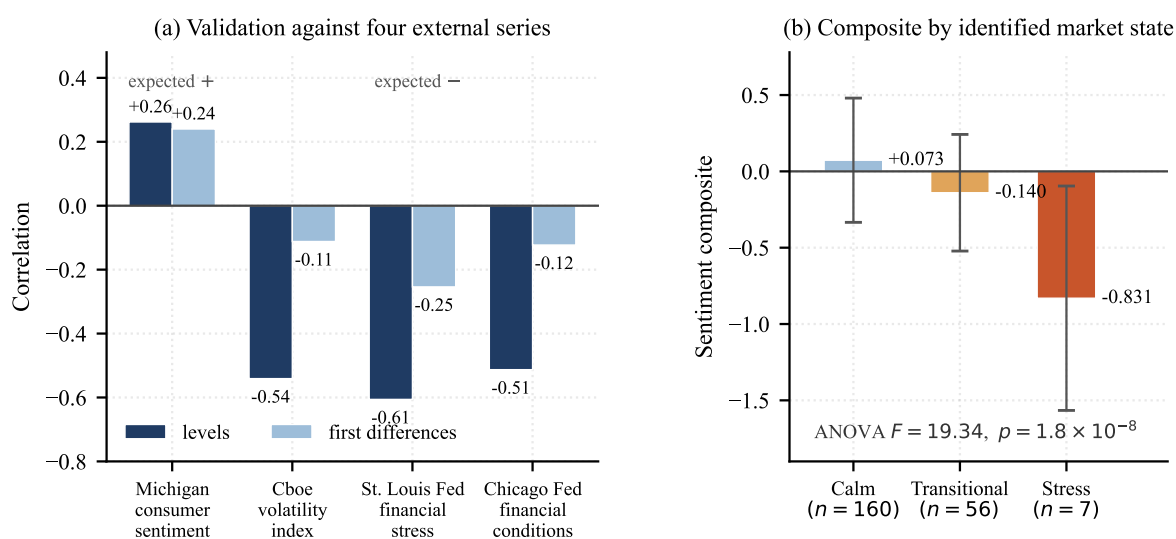
The recovered graph has a density of 0.850 and a maximum causal score of 1.000, indicating strong directed interdependence. Its most interpretable feature is that RELIANCE acts as a systematic causal hub: it exerts strong directed influence on HDFCBANK (0.913), TITAN (0.849) and TCS (0.650) while receiving markedly weaker influence in return. The decisive evidence for the approach is the comparison between correlation and causality — pairs exist in this universe, RELIANCE on TITAN being the clearest, for which the directed causal influence is high while the linear correlation is low, so a correlation-based selector would have discarded the relationship. Transparency and accuracy are therefore not in tension here, because the causal constraint retains information the correlational alternative destroys.

Performance over the study window is reported in full. The strategy attained a Sharpe ratio of 0.377 with a win rate of 48.8 per cent, and underperformed buy-and-hold by 8.95 per cent during a strong bull market. A win rate below one half combined with a positive Sharpe ratio indicates asymmetric profit capture, with winning trades larger than losing ones; transaction costs and signal lag account for the nominal shortfall, and the framework is a capital-preservation instrument rather than a return-maximising one. Peak volatility of 77.67 per cent during the 2024 Union Budget window shows the limit of the debiasing layer, whose stationarity assumption fails exactly when policy shocks change the noise process — a diagnostic rather than fatal result, since it identifies non-stationary debiasing as a specific, addressable weakness.

### 5.4.3 Sentiment noise and sentiment–price temporal dynamics

The second half of the objective concerns how noisy the sentiment channel is and how sentiment shifts relate in time to price movements. Three instruments were used.

**Construct validation against external evidence.** Naming a series a sentiment measure is a claim testable independently of whether it makes money. The market-derived composite [91] was compared against four independently published series, none of which enters the framework at any stage: a household survey of consumer sentiment, an implied volatility index, a financial stress index and a financial conditions index (Figure 5.2(a)). Every correlation carries the expected sign — positive against the survey, negative against the three stress and volatility measures — and every one is significant in first differences at the five per cent level. The survey comparison is the informative one, because the two series share no inputs: a correlation of  $+0.240$  between monthly changes in a composite built from exchange turnover and monthly changes in what households report to an interviewer cannot be a mechanical artefact of shared construction. The sign is correct in all sixteen series-by-sub-period combinations, with magnitudes largest in crisis periods, which is the expected pattern for a stress-sensitive measure.



**Figure 5.2:** Sentiment diagnostics. (a) Correlation of the sentiment composite with four independently published series in levels and first differences. (b) The composite evaluated by identified market state; the states are labelled by fitted volatility and the composite plays no part in fitting them.

**Separation across market states.** If the identified states are states of sentiment as well as of volatility, the composite should differ across them. Figure 5.2(b) shows a monotone and steep ordering:  $+0.073$  in the calm state,  $-0.140$  in the transitional state and  $-0.831$  in the stress state. One-way analysis of variance gives  $F = 19.34$  with  $p = 1.8 \times 10^{-8}$ , and the rank-based Kruskal–Wallis test, safer given only seven observations in the stress state, gives  $H = 29.54$  with  $p = 3.9 \times 10^{-7}$ . Since the mixture model is fitted on realised volatility and short-horizon momentum, and the composite is not among its inputs, this separation on a withheld variable justifies describing the states as sentiment-and-volatility states rather than volatility states alone.

**Noise and lead–lag structure.** In [P3], the temporal overlay of sentiment against price shows that strong sentiment spikes frequently precede major price changes, but that the mapping from magnitude to direction is not simple: sustained positive sentiment accompanies gradual advances, whereas sharp negative spikes accompany abrupt declines. The sentiment distribution is approximately symmetric about neutral with moderate tails, indicating that the extraction pipeline captures genuine variation rather than producing a biased categorical output. In [P5], the predictive content of the validated composite, measured as a cross-sectional information coefficient, is indistinguishable from zero or significantly negative depending on the period. Keeping these two results apart is essential: a composite can measure sentiment accurately and still fail to forecast returns.

#### 5.4.4 Summary and validation of Research Objective 3

Transparency here derives from directed causal structure rather than from post-hoc attribution alone, and multi-scale temporal encoding makes the horizon of each signal explicit. The framework recovers an interpretable causal graph of density 0.850 with an identifiable hub and demonstrates that causal influence and linear correlation can diverge, which is the justification for causal feature selection. Sentiment noise was quantified rather than assumed: the construct was validated against four external series with the correct sign in all sixteen series-by-period combinations and separates market states at  $p = 1.8 \times 10^{-8}$ , yet carries no reliable directional information — a distinction reported in full because it constrains how sentiment should be used downstream. [P2] supplies causal discovery, multi-granularity encoding and credibilistic allocation; [P3] and [P5] supply the sentiment-noise and temporal-dynamics evidence.

#### 5.5 Proposed Methodology for Research Objective 4

**RO4:** *To evaluate and validate the result and practical application of the proposed model across different stock markets, exploring transfer learning techniques for cross-market applications and validating the model against benchmark datasets and real-world market data.*

##### 5.5.1 Design of the decision-theoretic allocation framework

The framework converts a fused signal into an executable portfolio in a way that keeps two decisions separate: what to hold relative to everything else, and how much total risk to carry. That separation is what makes each layer independently testable. In **Stage 1**, nine causal features per asset per session are standardised cross-sectionally into a technical composite  $T_{i,t}$  and a sentiment composite  $M_{i,t}$ , so that a value expresses how an asset ranks against its peers on that day rather than against its own history; this also removes the common market factor, preventing a market-wide move from registering as a signal on every asset at once. In **Stage 2** the market state  $r_t$  is identified by a three-component Gaussian mixture over rolling volatility and momentum, refitted at every rebalance on an expanding window of strictly earlier sessions, with components anchored by ascending fitted volatility into calm, transitional and stress states so that labels remain comparable across refits; the composites are then fused with state-dependent weights and a confidence term  $c_{i,t}$  that discounts assets whose composites disagree, and the fused score is placed on a return scale by information-ratio scaling [92]:

$$z_{i,t} = c_{i,t} (w_{r_t}^T T_{i,t} + w_{r_t}^M M_{i,t}), \quad \alpha_{i,t} = \text{IC} \cdot \sigma_{i,t} \cdot z_{i,t}. \quad (5.2)$$

**Stage 3** splits the allocation in two. Relative composition solves a constrained quadratic program for the fully invested risky sleeve, with covariance shrinkage [82], a 25 per cent position cap and a turnover penalty annualising a one-way cost of ten basis points,

$$\tilde{w}_t = \arg \max_w \alpha_t^\top w - \lambda w^\top \Sigma_t w - \kappa \|w - w_{t-1}\|_1 \quad \text{s.t.} \quad w \geq 0, w \leq 0.25, \mathbf{1}^\top w = 1, \quad (5.3)$$

after which total risk is set by the regime budget  $w_t^* = k_{r_t} \tilde{w}_t$  with  $k = \{1.00, 0.75, 0.50\}$  expressed as a fraction of the sleeve’s own volatility and the residual held in cash. Scaling is one-sided, so the framework de-risks in disturbed states but never levers up, and any performance difference is attributable to risk reduction rather than to leverage. **Stage 4** evaluates the result by 21-session walk-forward with 10 bp costs, an 18-cell specification sweep, matched-exposure ablation, four stress episodes, a 665-session holdout and a secondary market universe, with Newey–West and stationary block bootstrap inference.

Two implementation details proved consequential and generalise beyond this framework. Mixture component indices permute freely between refits, so labels must be anchored; omitting the anchoring step produces no error, only a silently scrambled regime series. And the risk budget must be expressed as a fraction of the portfolio’s own volatility rather than as an absolute target, because absolute targets calibrated for concentrated equity portfolios are never reached by a diversified sleeve and leave the regime layer present in the specification but inert — here binding in one rebalance out of 224.

### 5.5.2 Results and discussion for Research Objective 4

Table 5.4 consolidates the evaluation across the primary universe, the holdout, the specification sweep and the secondary market.

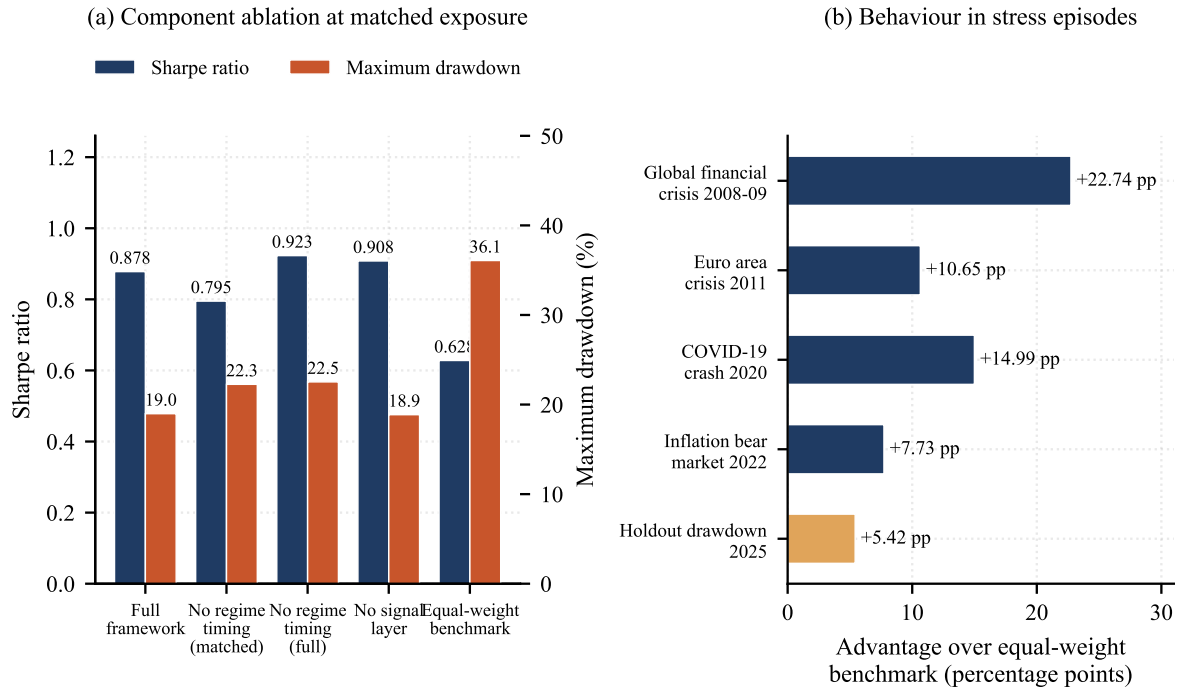
**Table 5.4:** Portfolio-level evaluation. Benchmarks are rebalanced daily and charged no transaction costs, which is deliberately unfavourable to the proposed framework.

| Configuration                                                                                                          | CAGR     | Vol.   | Sharpe        | Max DD          | Calmar |
|------------------------------------------------------------------------------------------------------------------------|----------|--------|---------------|-----------------|--------|
| <i>Primary universe: ten multi-asset ETFs, Jan 2008 – Aug 2026, 4,689 sessions</i>                                     |          |        |               |                 |        |
| Proposed regime-conditional framework                                                                                  | 5.65%    | 6.44%  | <b>0.878</b>  | –18.98%         | 0.298  |
| Equal-weight benchmark                                                                                                 | 8.45%    | 13.45% | 0.628         | –36.08%         | 0.234  |
| Inverse-volatility benchmark                                                                                           | 7.46%    | 8.78%  | 0.849         | –24.09%         | 0.310  |
| Equal-weight at strategy volatility                                                                                    | 4.19%    | 6.44%  | 0.651         | –18.69%         | 0.224  |
| <i>Out-of-sample holdout: 665 sessions, Jan 2024 – Aug 2026</i>                                                        |          |        |               |                 |        |
| Proposed framework                                                                                                     | 12.32%   | 7.63%  | <b>1.614</b>  | –5.48%          | 2.246  |
| Equal-weight benchmark                                                                                                 | 15.44%   | 11.10% | 1.391         | –10.13%         | 1.524  |
| Inverse-volatility benchmark                                                                                           | 11.89%   | 8.51%  | 1.397         | –6.70%          | 1.775  |
| Equal-weight at strategy volatility                                                                                    | 7.28%    | 5.32%  | 1.370         | –7.04%          | 1.473  |
| <i>Specification sweep: 3 seeds <math>\times</math> 2 covariance windows <math>\times</math> 3 rebalance intervals</i> |          |        |               |                 |        |
| Range across all 18 cells                                                                                              | —        | —      | 0.813 – 0.956 | –21.1% – –19.0% | —      |
| Cells beating the benchmark on both measures                                                                           | 18 of 18 |        |               |                 |        |
| <i>Secondary universe: ten mega-capitalisation equities, 2019 – 2026 (cross-market transfer)</i>                       |          |        |               |                 |        |
| Proposed framework                                                                                                     | 17.02%   | 14.87% | 1.145         | –20.45%         | 0.832  |
| Equal-weight benchmark                                                                                                 | 33.52%   | 24.45% | <b>1.371</b>  | –33.63%         | 0.997  |

On the primary universe the framework raises the Sharpe ratio from 0.628 to 0.878, a gain of 40 per cent, and reduces maximum drawdown from –36.08 to –18.98 per cent, a reduction of 47 per cent, while earning a lower compound return of 5.65 against 8.45 per cent per annum. It is a risk-reduction instrument whose return sacrifice is smaller than its risk reduction, and it is presented as such. Against the volatility-matched benchmark it retains its Sharpe advantage but loses marginally on drawdown, which bounds the drawdown claim precisely: the framework reduces drawdown relative to an unlevered diversified benchmark, not relative to every portfolio held at the same volatility. The holdout block is the strongest single piece of evidence, because every specification decision — universe, feature set, number of regimes, risk budget, rebalancing interval and cost assumption — was fixed before any session in it was examined; on those 665 sessions the framework records a higher Sharpe ratio and a smaller drawdown than all three benchmarks, and two of those comparisons reverse the in-sample ordering. Three qualifications are stated alongside: 665 sessions cannot settle the question on their own, the period contains no episode comparable to 2008 or 2020, and the framework again earns less in absolute terms.

The ablation answers the objective’s central question. Holding average exposure fixed at 92.1 per cent, allocating that exposure according to the identified regime rather than uniformly raises the Sharpe ratio from 0.795 to 0.878 and improves maximum drawdown from –22.26 to –18.98 per cent, with no significant difference in mean return ( $p = 0.695$ ). That combination — both measures improve, mean return does not change — is the signature of a genuine risk-timing effect rather than of simply holding less, and it is available only because the control was constructed at matched risk. The cost of that timing is also measured: against a fully invested variant the framework gives up about 1.3 percentage points of annual return, significantly so ( $p = 0.016$ ), in exchange for 3.55 points of maximum drawdown.

The same ablation returns a null on the signal fusion layer. Removing it entirely, so that the optimiser becomes a constrained minimum-variance problem, moves the Sharpe ratio from 0.878 to 0.908 and maximum drawdown from –18.98 to –18.87 per cent; both movements favour removal and neither is significant ( $p = 0.650$ ). The information coefficients explain why: the technical, sentiment and fused composites are all significantly *negatively* related to next-session cross-sectional returns over the full sample (IC = –0.030, –0.026 and –0.033; Newey–West  $t = -2.14, -2.26, -2.42$ ), and the state-dependent



**Figure 5.3:** (a) Component ablation. The matched-exposure control carries the same average invested exposure as the full framework but applies it without regard to the regime, so that only the timing of risk differs. (b) Cumulative advantage over the benchmark in four stress episodes chosen on documented market history, plus the deepest benchmark drawdown in the holdout.

weighting hypothesis — that sentiment becomes relatively more informative under stress — is not supported, with point estimates running the other way and the calm-to-stress contrast insignificant ( $t = -1.46$ ,  $p = 0.145$ ). The relationship is not stable enough to exploit by inversion either:  $-0.039$  before 2024 and  $+0.009$  afterwards. This null is retained in full. Component-level negative results are rarely published, which is one reason architectures accumulate layers whose contribution has never been isolated; reporting which layer works is more useful than an aggregate number that conceals the distinction.

Behaviour under stress is consistent across four episodes selected before the analysis on documented market history rather than by inspection of results. The framework outperforms in all four, by between 7.7 and 22.7 percentage points, and by a further 5.4 points in the deepest benchmark drawdown of the holdout. Its average return on the worst 5 per cent of benchmark sessions is  $-0.59$  per cent against  $-2.01$  per cent, a cushion of 1.42 points per session, and downside capture of 0.38 against upside capture of 0.41 gives a capture ratio of 0.93: the framework participates less in both directions in roughly equal proportion, so its advantage comes from carrying less risk in adverse states rather than from directional skill. On statistical significance the report is equally direct. The Sharpe difference of  $+0.250$  carries a 95 per cent bootstrap interval of  $[-0.102, 0.574]$ , which includes zero, and the shortfall in mean return is not significant ( $p = 0.171$ ). Eighteen years of daily data contain only four independent observations of the events that matter for a drawdown claim, and no amount of daily sampling converts that into statistical power. What is robust is consistency, not the point estimate: the Sharpe advantage appears in 18 of 18 specifications, the drawdown advantage in 18 of 18 specifications and 18 of 19 calendar years, and the outperformance in all four stress episodes. Robustness and significance are different properties and are not conflated here.

### 5.5.3 Cross-market transfer

Transfer was tested by running the framework without modification on a second universe of ten mega-capitalisation United States equities over 2019–2026. The outcome is mixed and is reported as such. The framework loses on risk-adjusted return, 1.145 against 1.371, and gives up 15.09 percentage points of annual return, significantly so ( $t = -2.791$ ,  $p = 0.005$ ). Its defensive behaviour nevertheless survives intact: maximum drawdown improves from  $-33.63$  to  $-20.45$  per cent, it wins on drawdown in eight of eight calendar years and in all 36 specification cells examined on that universe, and it outperforms in both

stress episodes the window contains, by 12.50 percentage points in the 2020 crash and 12.60 points in the 2022 bear market. Critically, the regime layer continues to behave as designed: against the matched-exposure control it improves both the Sharpe ratio and the drawdown, so the risk-timing effect reproduces in a universe where the framework loses overall.

One point of scope should be stated plainly rather than glossed. The objective refers to *transfer learning*; what has been carried out here is *direct transfer*. The framework, its feature definitions and its fitted procedure were applied to the second universe without modification and without any adaptation step — the stricter test of whether the mechanism itself generalises, since no parameter was fine-tuned on the target market, no representation was pre-trained and re-used, and no domain-adaptation objective was optimised. The result is therefore a lower bound on cross-market performance, and adaptation-based transfer is identified as open work in Section 7.2 rather than claimed here. The explanation is a property of the universe rather than of the framework: ten mega-capitalisation equities are close to a single factor bet, with high correlations, no low-volatility asset to rotate into, and a risky sleeve carrying roughly 15 per cent annualised volatility against 6 per cent in the multi-asset case, so de-risking can only move capital to cash and the framework systematically forgoes return during an exceptional advance. The transferability condition is therefore explicit and testable in advance: the framework requires a universe with genuine dispersion in asset volatility. Complementary cross-market evidence at sectoral granularity comes from the ten-sector evaluation of [P1] and the sector-level regime behaviour of [P2].

#### 5.5.4 Summary and validation of Research Objective 4

The framework was evaluated as a decision system rather than as a predictor. It improves risk-adjusted return and materially reduces drawdown against an unlevered diversified benchmark, sustains that behaviour on a holdout that postdates every design decision, and does so consistently across every specification examined. Ablation at matched exposure locates the gain in regime-conditional risk timing and returns a null on the signal fusion layer, and cross-market transfer establishes a precise boundary condition for applicability. The limits of the evidence — an insignificant Sharpe improvement, few independent stress events, a lower absolute return — are stated explicitly. [P5] supplies the framework, the eighteen-year evaluation, the specification sweep, the ablation, the holdout and the cross-market test; [P1] and [P2] supply supporting sectoral and regime-level evidence.

#### 5.6 Integration of the Individual Contributions

The four contributions are not four independent studies with a shared subject; each was produced in response to a limitation exposed by the one before it. Contribution 1 established that fusing four sources improves prediction and that the value of each source is state-dependent, which raised the question of how to weight a source whose reliability changes — answered by the uncertainty-weighted formulation. Contribution 2 isolated the language-model sentiment channel and found it the most important single feature yet delivering only a bounded increment in raw accuracy, which raised the question of whether the limitation lies in the measurement or in the horizon. Contribution 3 answered by separating the two: the construct was validated externally and found sound, while its predictive content was measured separately and found weak, so the limitation is not one of measurement. Contribution 4 followed directly — if the signal channel is bounded, the framework must earn its value at the decision layer, and the ablation confirms exactly this: risk timing by market state works, and the signal layer does not. The final system is therefore not the one originally envisaged, and it is better supported for that reason.

## Chapter 6: Results Discussion and Research Significance

### 6.1 Consolidated Outcomes Against the Objectives

Table 6.1 summarises what each objective produced and the status of the evidence behind it. The final column is stated deliberately, because the strength of a result and the confidence that can be placed in it are separate properties and are treated separately throughout this work.

**Table 6.1:** Objective-wise consolidated outcomes and the status of the supporting evidence

| Obj. | Principal outcome                                                                        | Key quantitative evidence                                                                                                                                                                                                         | Status and limits                                                                                                     |
|------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| RO1  | Four-stream fusion model with adaptive, then uncertainty-conditioned, source weighting   | MAE 0.018, $R^2$ 0.988, DA 84.7% against 76.3% for a fixed-weight ensemble on identical inputs; both textual sources contribute 10.9% of $R^2$ superadditively; dynamic weighting adds 22.7% Sharpe and removes 18.3% of drawdown | Established. Index-level, one-day horizon; macro interpolation understates announcement-day effects                   |
| RO2  | Hybrid LLM sentiment and indicator framework in a reproducible environment               | Sentiment is the highest-gain feature; correlation with price 0.04, with momentum 0.74; accuracy 47.72% against a 46.43% baseline mean; highest prediction confidence                                                             | Established but bounded. All models near the random classifier at a single-name daily horizon                         |
| RO3  | Causal, multi-granularity transparency and a measured account of sentiment noise         | Causal graph density 0.850 with an identifiable hub; correct external sign in 16 of 16 series-by-period tests; market states separated at $p = 1.8 \times 10^{-8}$                                                                | Established for construct validity. Predictive content of the composite is separately measured and is not significant |
| RO4  | Regime-conditional decision framework, ablated at matched risk and tested across markets | Sharpe 0.878 against 0.628; drawdown $-18.98\%$ against $-36.08\%$ ; wins in 18/18 specifications, 18/19 years, 4/4 stress episodes; holdout Sharpe 1.614                                                                         | Established as consistent, not as statistically significant. Bootstrap interval $[-0.102, 0.574]$ includes zero       |

Read as a ledger the pattern is consistent: adaptive fusion beats fixed-weight combination, but index-level accuracy does not transfer to single-name daily direction; language-model sentiment is orthogonal and highest-importance, but does not overturn market efficiency at a daily horizon; the sentiment construct is validated, but a validated construct does not thereby forecast returns; regime risk timing improves both Sharpe ratio and drawdown at matched exposure, but the Sharpe gain is not statistically significant on eighteen years of data; and the defensive behaviour transfers to a second universe, though the framework does not add value in a concentrated single-factor one. These bounding statements are not shortcomings but the boundary of the contribution, and stating them is what makes the positive claims credible.

### 6.2 Reconciling the Divergent Performance Across Studies

A reader comparing the objectives will notice an apparent contradiction: Objective 1 reports 84.7 per cent directional accuracy while Objective 2 reports 47.72 per cent. The two numbers are not in conflict, and understanding why is central to the significance of this research. They answer different questions on different objects. Objective 1 predicts the next-day level of a diversified index from four fused sources; an index aggregates hundreds of constituents, so idiosyncratic noise largely cancels and the remaining signal is dominated by persistent market-wide state, which fusion captures well. Objective 2 predicts the next-day direction of an individual equity, which carries its full idiosyncratic component, has close to balanced classes and offers no aggregation to average the noise away. Predictability is therefore expected to be far lower, and this is what the empirical asset pricing literature reports [94, 93].

Three implications follow, and together they define the trajectory of the research programme. First, an accuracy figure is meaningless without its object and horizon. Second, the level at which a system is

applied should match where the signal actually lives: aggregate state is predictable, single-name daily direction is largely not. Third, and most consequentially, if the signal channel is weak at the level where decisions are taken, value must be created at the decision layer rather than the prediction layer. Objective 4 tests exactly that, and the matched-exposure ablation confirms it. The apparent contradiction between the two accuracy figures is therefore not an inconsistency but the finding that organises the work.

### 6.3 Research Significance

**Methodological significance.** The work advances multi-source fusion from a question of *what* to combine to a question of *how* to combine under changing reliability. Expressing fusion weights as a function of measured, time-varying predictive uncertainty [5, 104] gives a source-weighting rule that self-corrects within days rather than at the next retraining cycle, and distribution-aware normalisation prevents a heavy-tailed input from dominating the fused representation. Separating the allocation decision into relative composition and total risk is a second contribution: it makes each layer independently falsifiable, and without it the central ablation of this thesis could not have been constructed at all.

**Evaluation significance.** Three practices used here are uncommon in this literature and are the most transferable part of the work: ablation at matched risk, which separates the timing of risk from its level, since comparing a de-risked strategy against a fully invested benchmark confounds the two and credits a framework for merely holding less; disclosure of the entire specification sweep rather than its best cell, the minimum required once multiple testing is taken seriously [95, 96, 97]; and separation of construct validity from predictive usefulness, the sentiment composite being shown to measure sentiment and, separately, shown not to forecast returns. Conflating those questions is a recurring error, and keeping them apart changes what a null result means.

**Engineering significance.** Two failure modes were diagnosed that generalise beyond this framework. The first is scale mismatch in signal-to-allocation conversion: passing an unscaled indicator into a mean-variance objective lets the ratio of units between the linear and quadratic terms, rather than the information in the signal, determine the solution, producing a degenerate single-asset portfolio; information-ratio scaling [92] removes it. The second is an inert risk layer: an absolute volatility target calibrated for a concentrated equity portfolio binds in one rebalance out of 224 on a diversified universe, so the layer is present, consumes computation and does nothing, whereas a relative budget is scale-free and remains active. Mixture-model label anchoring has the same character — omitting it produces no error message, only a silently scrambled regime series.

**Practical significance.** For a practitioner the results support a specific and limited prescription: fuse sources but weight them by measured reliability rather than by historical average importance; use language-model sentiment for what it demonstrably provides — orthogonal information and better-calibrated confidence — rather than for a directional edge it does not provide at a daily horizon; prefer causally selected features where the structure is recoverable, because they survive regime change better and can be audited; implement a regime risk budget in relative rather than absolute form, evaluate it against a volatility-matched benchmark, and expect it to help only where the universe contains genuine dispersion in asset volatility; and size positions conservatively, in the fractional form the growth-optimal criterion recommends once estimation error is admitted [80]. For the Indian market specifically the measured weights matter: macroeconomic indicators carry 25 per cent of average attention, sentiment sources jointly carry 40 per cent with the social media share rising sharply at market extremes, and sectoral predictability varies systematically with information availability and factor exposure — measured properties of that information environment rather than assumptions imported from elsewhere.

## Chapter 7: Conclusion and Future Scope

### 7.1 Conclusion

This research set out to integrate multi-source data with sentiment analysis and language models in order to improve stock market decision making, and to determine which parts of such an integration actually contribute to the outcome. Both aims are met, though not entirely in the direction originally anticipated, and the divergence is itself the most useful result.

A four-stream fusion model integrating financial metrics, news sentiment, social media sentiment and macroeconomic indicators was developed and evaluated on ten years of Indian market data across ten sectoral indices. It attains a normalised mean absolute error of 0.018,  $R^2$  of 0.988 and directional accuracy of 84.7 per cent against 76.3 per cent for a fixed-weight ensemble over identical inputs, with stability confirmed by walk-forward evaluation at  $0.0016 \pm 0.0004$  normalised error; removing both textual sources costs 10.9 per cent of explained variance, exceeding the sum of their individual removals — the quantitative signature of complementary information. The fusion rule was then advanced from learned average relevance to measured instantaneous reliability through a Bayesian formulation in which weights are obtained by a softmax over negative predictive uncertainties, improving the Sharpe ratio by 22.7 per cent and reducing maximum drawdown by 18.3 per cent against the strongest static-fusion baseline while the same experts fused statically add nothing.

A hybrid framework coupling a transformer language model with fifteen conventional technical indicators was implemented in a reproducible computational environment and evaluated under a strict temporal protocol. Language-model sentiment emerged as the highest-gain feature in the fused space, correlating 0.74 with momentum yet only 0.04 with price, establishing that it supplies orthogonal information. The gain in raw accuracy is modest — 47.72 per cent against a baseline mean of 46.43 per cent — and all models operate close to the random classifier at a single-name daily horizon. That limit is reported rather than concealed, and it redirected the research toward the decision layer. Transparency was then established structurally: a directed causal adjacency matrix recovered by conditional independence testing yields a graph of density 0.850 with an identifiable causal hub and demonstrates that strong directed influence can coexist with weak linear correlation — the precise justification for causal rather than correlational feature selection. Sentiment noise was quantified rather than assumed: the composite carries the expected sign against four independently published series in all sixteen series-by-sub-period combinations and separates the identified market states at  $p = 1.8 \times 10^{-8}$ , while its predictive content, measured separately, is not significant. Keeping construct validity and forecasting power apart is a contribution in its own right.

The decision framework was evaluated over eighteen years on ten liquid multi-asset instruments under a walk-forward protocol with transaction costs. It attains a Sharpe ratio of 0.878 against 0.628 for an equal-weighted benchmark and reduces maximum drawdown from  $-36.1$  to  $-19.0$  per cent, outperforming in all four stress episodes, in 18 of 19 calendar years on drawdown and in all 18 specifications examined; on a 665-session holdout that postdates every design decision it records a higher Sharpe ratio and a smaller drawdown than all three benchmarks. Ablation at matched average exposure locates the source precisely: timing risk by identified market state improves both the Sharpe ratio and the drawdown with no significant change in mean return, whereas the signal fusion layer contributes nothing detectable. Cross-market transfer reproduces the defensive behaviour in a concentrated mega-capitalisation universe while losing on risk-adjusted return, bounding applicability to universes with genuine dispersion in asset volatility.

The overall conclusion is precise rather than expansive. Multi-source integration and language-model sentiment demonstrably improve the description of the market, and causal structure demonstrably improves the transparency of that description. At the horizon and granularity at which decisions are actually taken, however, the measurable value of the system arises at the allocation layer, through regime-conditional risk timing, rather than at the signal layer. A framework whose every layer has been separately tested, and whose applicability has been bounded by evidence, is a more useful engineering artefact than one whose aggregate performance conceals which of its components is doing the work.

## 7.2 Future Scope

- **Textual sentiment in a universe with instrument-level news coverage.** The null reported for the market-derived composite is a null about that proxy, not about textual sentiment. Applying the fusion hypothesis where a time-stamped, instrument-level news corpus exists would convert a proxy null into a genuine test, and would connect the language-model pipeline of Objective 2 directly to the allocation layer of Objective 4 [74, 14].
- **Bayesian neural networks for richer uncertainty.** The present uncertainty estimate is a rolling error variance, which captures aleatoric variation. Replacing it with proper Bayesian neural expert models would additionally expose epistemic uncertainty, producing a confidence measure that distinguishes a noisy environment from an unfamiliar one [103, 104, 16].
- **Non-stationary debiasing and scalable causal discovery.** The stationarity assumption in the multi-scale debiasing layer fails at exactly the moments that matter, as the 77.67 per cent peak volatility during the Union Budget window shows; an adaptive formulation whose noise model is itself state-dependent is the direct remedy. In parallel, extending conditional independence testing from five instruments to the full fifty-constituent index requires an algorithm whose cost does not grow quadratically, which is a prerequisite for causal transparency at production scale [60, 102].
- **Adaptation-based cross-market transfer.** The cross-market evidence reported here comes from applying the framework to a second universe without modification, which tests whether the mechanism generalises but forgoes any benefit from adapting to the target market. The natural extension is genuine transfer learning: pre-training the regime model and the source-reliability estimates on a data-rich market and fine-tuning them on a thinner one, so that an emerging-market deployment inherits structure learned elsewhere instead of starting from its own limited history.
- **Learned risk budgets and explicit state persistence.** The regime multipliers are fixed at 1.00, 0.75 and 0.50; learning them — reinforcement learning being the natural candidate given the sequential structure [76] — would adapt the budget to the universe, subject to the overfitting controls this work argues are mandatory [95]. Replacing the Gaussian mixture with a hidden Markov or jump formulation would represent regime persistence directly rather than recovering it incidentally [88, 85].
- **Regional-language and higher-frequency sentiment.** Sentiment extraction is currently English-only and daily. A large share of Indian retail participants read financial news in regional languages, and intraday resolution would expose incorporation dynamics that daily aggregation averages away.
- **Deployment engineering.** Inference cost is negligible; the binding constraints are the latency of retrieving and scoring text at scale, feed rate limits, coordinated manipulation of social signals, transaction costs and compliance with algorithmic trading regulation. These are treated as first-class engineering problems rather than as afterthoughts.

## Chapter 8: Organisation of the Thesis

The thesis is organised into nine chapters.

**Chapter I — Introduction.** Introduces stock market decision making as an information-integration problem, describes the structure of the Indian equity market, and states the motivation for combining quantitative, textual and macroeconomic evidence within a single system.

**Chapter II — Background and Preliminaries.** Presents the technical foundations: technical indicator construction, sentiment extraction from lexicon methods through to transformer language models, fusion strategies, regime modelling, and the evaluation metrics and statistical procedures used throughout the thesis.

**Chapter III — Literature Survey and Research Gaps.** Reviews multi-source fusion, language-model sentiment analysis, explainable and causal financial modelling, and decision-level allocation, and derives the five research gaps that define the scope of the work.

**Chapter IV — Multi-Source Data Integration Framework.** Develops the Multi-Source Fusion Network: data acquisition and temporal alignment, modality-specific encoders, cross-modal attention, the Bi-LSTM predictor, the experimental setup, the sectoral and ablation results, and the walk-forward robustness analysis.

**Chapter V — Dynamic Uncertainty-Weighted Fusion.** Develops the Bayesian formulation of the fusion rule: expert models per source, time-varying uncertainty estimation, softmax weighting over negative uncertainties, distribution-aware normalisation, and comparison against static early, late and attention-based fusion baselines.

**Chapter VI — Hybrid Language Model and Indicator Framework.** Develops the sentiment extraction pipeline, feature-level fusion with technical indicators, model training and comparison, feature-importance and sentiment-regime analysis, confidence calibration and the simulated trading evaluation.

**Chapter VII — Explainable and Causal Framework.** Develops causal graph discovery, multi-granularity temporal memory encoding and credibilistic optimisation; reports the recovered causal structure, the contrast between correlation and causality, the sentiment noise diagnostics and the sentiment-to-price temporal dynamics.

**Chapter VIII — Decision-Theoretic Allocation and Cross-Market Validation.** Develops regime identification, constrained portfolio optimisation and the regime risk budget; reports the eighteen-year evaluation, the specification sweep, the matched-exposure ablation, the stress-episode analysis, the holdout and the cross-market transfer test.

**Chapter IX — Conclusion and Future Scope.** Consolidates the findings across the four objectives, states the limits of the evidence, and sets out the future research directions.

## Publications on PhD Work

### International Journals

- [P3] A. Pardeshi and S. Deshmukh, “Fusing LLM-Extracted News Sentiment and Market Data for Enhanced Stock Movement Prediction,” *Liberte Journal*, ISSN 0024-2020, vol. 14, no. 6, pp. 424–439, 2026. **(Published)**  
Indexing: [ indexing, e.g. Scopus / UGC-CARE Group II / not indexed ]  
[ article URL or DOI ]
- [P4] A. Pardeshi and S. Deshmukh, “Dynamic Uncertainty-Weighted Fusion for Multi-Source Financial Data: A Bayesian Framework for Time-Varying Source Reliability,” *International Journal of Computer Information Systems and Industrial Management Applications (IJCISIM)*. **(Communicated, [ submission month and year ])**  
[ submission acknowledgement URL or manuscript ID, if any ]
- [P5] A. Pardeshi and S. Deshmukh, “From Signal Fusion to Asset Allocation: A Decision-Theoretic Model for Portfolio Construction Under Regime-Based Sentiment and Volatility,” *International Journal of Applied Decision Sciences* (Inderscience), manuscript IJADS-312370. **(Under review — revised manuscript submitted after a major-revision decision)**  
[ Inderscience submission URL, if you wish to include it ]

### International Conferences

- [P1] A. Pardeshi and S. Deshmukh, “A Multi-Source Data Fusion Framework for Enhanced Stock Market Prediction,” *2026 International Conference on Sustainability, Innovation and Technology (ICSIT)*, [ host institute, city, and conference dates ], IEEE, 2026, [ pages, or “in press” ].  
[ IEEE Xplore DOI or proceedings URL ]
- [P2] A. Pardeshi and S. Deshmukh, “A Dynamic-Causal Hybrid Framework for NIFTY50 Stock Decision Making: A Multi-Granularity Temporal Learning Approach,” [ full conference name, host institute, city and dates ]. **(Communicated)**  
[ submission acknowledgement URL or paper ID, if any ]
- [P6] A. Pardeshi and S. Deshmukh, “Deep Learning in Stock Market Forecasting: Comparative Insights and Future Directions,” *2025 International Conference on Emerging Trends in Industry 4.0 Technologies (ICETI4T)*, Navi Mumbai, India, IEEE, June 2025, [ pages ].  
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*Note:* The labels [P1]–[P6] are used throughout this report to refer to the above works. Publication status is as of the date of submission of this report.

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